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- Two-dimensional arrays
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12.2.1 One-dimensional array

This is the simplest form of an array. A one dimensional array represents values that are stored in a single row or in a single column.

Declaration

Syntax:

<data type><array_name> [<array_size>];

data_type declares the basic type of the array, which is the type of each element in the array.

array_name specifies the name with which the array will be referenced.

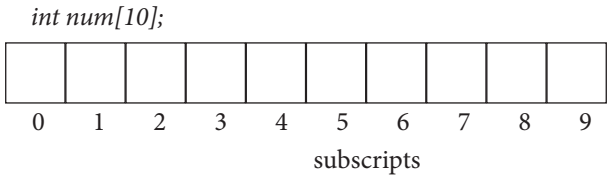
array_size defines how many elements the array will hold. Size should be specified with square brackets [ ].

Example:

int num[10];

In the above declaration, an array named "num" is declared with 10 elements (memory space to store 10 different values) as integer type.

To the above declaration, the compiler allocated 10 memory locations (boxes) in the common name "num" as given below



Each element (Memory box) has a unique index number starting from 0 which is known as "subscript". The subscript always starts with 0 and it should be an unsigned integer value. Each element of an array is referred by its name with subscript index within the square bracket. For example, num[3] refers to the 4th element in the array.

Some more array declarations with various data types:

char emp_name[25]; // character array named emp_name with size 25

float salary[20]; // floating-point array named salary with size 20

int a[5], b[10], c[15]; // multiple arrays are declared of type int

Memory representation of an one dimensional array

The amount of storage required to hold an array is directly related with type and size. The following figure shows the memory allocation of an array with five elements.